

IE 376, Fall 2025-2026 Homework #3

Due on December 17, 2025, Wednesday by 5:00 pm

Question 1: Read the paper by Spear and Bowen titled “Decoding the DNA of the Toyota Production System”.

- a) Explain (in your own words) the four rules that made Toyota successful
- b) What examples did you find in the paper to demonstrate Kaizen at Toyota?

Question 2: Killjoy Electrical produces surge protectors for the company industry. On one of the assembly lines in its plant Killjoy produces three-end-product models of surge protectors, product 1, 2, and 3. The data shown below are for the three products and their components.

End Product	1	2	3
Planned Quantity	3	2	2

These products consume components X, Y and Z based on the following table

	Component Item Usage/End Product		
End Product	X	Y	Z
1	1	1	0
2	0	1	1
3	1	1	1

Use the Toyota approach to schedule the production of these 3 products to minimize variation in component production.